Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. **Consider the matrices**

$$A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}.$$

a) **Verify that B is the inverse of A.**

By the definition of an inverse matrix, we must verify that $AB = BA = I_2$.

$$\begin{aligned} AB &= \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 6-5 & -15+15 \\ 2-2 & -5+6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \end{aligned}$$

Similarly,

$$\begin{aligned} BA &= \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 6-5 & 10-10 \\ -3+3 & -5+6 \end{bmatrix} = I_2. \end{aligned}$$

Therefore,

$$\boxed{A^{-1} = B}.$$

b) **Determine whether the matrix**

$$D = \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$$

is the inverse of A. Justify your answer.

Note: D is an arbitrary name that we are using for the matrix.

We verify whether $AD = I_2$.

$$AD = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & -5 \\ 0 & -1 \end{bmatrix} \neq I_2 \implies \boxed{D \neq A^{-1}}.$$

c) **Determine $(5A)^{-1}$ and $(A^T)^{-1}$.**

From part a), we know that $A^{-1} = B$.

Using the property

$$(\lambda A)^{-1} = \lambda^{-1} A^{-1}, \qquad \lambda \neq 0,$$

we obtain

$$\begin{aligned} (5A)^{-1} &= 5^{-1} A^{-1} \\ &= \frac{1}{5} B \\ &= \frac{1}{5} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} \frac{2}{5} & -1 \\ -\frac{1}{5} & \frac{3}{5} \end{bmatrix}}. \end{aligned}$$

Similarly, using the property

$$(A^T)^{-1} = (A^{-1})^T,$$

we obtain

$$\begin{aligned} (A^T)^{-1} &= (A^{-1})^T \\ &= B^T \\ &= \boxed{\begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}}. \end{aligned}$$

Further justification.

We now justify these properties using the definition of an inverse matrix.

1. Inverse of a scalar multiple.

Assume that A is invertible and $\lambda \neq 0$.

Consider the matrix

$$X = \lambda^{-1} A^{-1}.$$

We verify that

$$\begin{aligned} X(\lambda A) &= (\lambda^{-1} A^{-1})(\lambda A) \\ &= (\lambda^{-1} \lambda)(A^{-1} A) \\ &= 1 \cdot I_2 \\ &= I_2. \end{aligned}$$

Likewise,

$$\begin{aligned}(\lambda A)X &= (\lambda A)(\lambda^{-1}A^{-1}) \\ &= (\lambda\lambda^{-1})(AA^{-1}) \\ &= I_2.\end{aligned}$$

Hence, by the definition of inverse,

$$\boxed{(\lambda A)^{-1} = \lambda^{-1}A^{-1}}.$$

2. Inverse of the transpose.
Assume that A is invertible. By definition,

$$A^{-1}A = AA^{-1} = I_2.$$

Taking the transpose of $AA^{-1} = I_2$,

$$\begin{aligned}(AA^{-1})^T &= I_2^T \\ (A^{-1})^T A^T &= I_2.\end{aligned}$$

Likewise, taking the transpose of $A^{-1}A = I_2$,

$$\begin{aligned}(A^{-1}A)^T &= I_2^T \\ A^T(A^{-1})^T &= I_2.\end{aligned}$$

Thus,

$$(A^{-1})^T A^T = A^T(A^{-1})^T = I_2.$$

Therefore, by the definition of inverse,

$$\boxed{(A^T)^{-1} = (A^{-1})^T}.$$

d) **Calculate C^3. Is C invertible? If so, determine its inverse.**
First, we compute C^2:

$$\begin{aligned}C^2 &= CC \\ &= \begin{bmatrix}0 & -1\\ 1 & -1\end{bmatrix}\begin{bmatrix}0 & -1\\ 1 & -1\end{bmatrix} \\ &= \begin{bmatrix}-1 & 1\\ -1 & 0\end{bmatrix}.\end{aligned}$$

Then,

$$\begin{aligned}C^3 &= C^2C \\ &= \begin{bmatrix}-1 & 1\\ -1 & 0\end{bmatrix}\begin{bmatrix}0 & -1\\ 1 & -1\end{bmatrix} \\ &= \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = I_2.\end{aligned}$$

Since $C^3 = I_2$, we have

$$C^2C = CC^2 = I_2.$$

Therefore, C is invertible and

$$\boxed{C^{-1} = C^2 = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}}.$$

e) **Determine, with justification, whether AC is invertible. If so, determine $(AC)^{-1}$.**

From parts a) and d), both A and C are invertible.

By the inverse-of-a-product property, AC is invertible and

$$(AC)^{-1} = C^{-1}A^{-1}.$$

Substituting the previously obtained inverses,

$$\begin{aligned} (AC)^{-1} &= C^{-1}B \\ &= \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} -3 & 8 \\ -2 & 5 \end{bmatrix}}. \end{aligned}$$

Therefore, AC is invertible.

We may verify the result directly:

$$AC = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix},$$

and

$$\begin{aligned} (AC)(AC)^{-1} &= \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} -3 & 8 \\ -2 & 5 \end{bmatrix} \\ &= I_2. \end{aligned}$$